COVID-19 Case Analysis Report

Generated: March 27, 2026

EXECUTIVE SUMMARY

HealthGuard Analytics Pvt. Ltd. conducted a comprehensive analysis of infectious disease case data to support public health decision-making. This report presents findings from 5,000 patient cases during the early phase of the outbreak (January 2020 - March 2020).

KEY FINDINGS

1. DEMOGRAPHIC PATTERNS (Who is getting infected?)

- Average Patient Age: 44.8 years
- Age Range: 1 to 90 years
- Most vulnerable age group: 45+ years

2. INFECTION PATTERNS (How are infections spreading?)

- Primary Infection Source: Contact with confirmed case (34.8%)
- Average Contacts per Patient: 5.0 individuals

3. RECOVERY TRENDS (What are the recovery patterns?)

- Overall Recovery Rate: 68.4% (3,422 patients)
- Average Recovery Time: 12.8 days
- Median Recovery Time: 12 days

4. REGIONAL IMPACT (Which regions are most affected?)

- Most Affected Region: Maharashtra (348 cases - 7.0%)
- Second: New York (347 cases - 6.9%)
- Third: Sao Paulo (346 cases - 6.9%)

5. PATIENT OUTCOMES

- Released (Recovered): 3,422 patients (68.4%)
- Isolated (Active): 1,019 patients (20.4%)
- Deceased: 559 patients (11.2%)

RECOMMENDATIONS

1. SCREENING PRIORITIZATION:

Focus screening efforts on the 45+ age group, as they represent a significant portion of cases.

2. CONTACT TRACING:

Implement aggressive contact tracing for individuals with high contact numbers (more than 10 contacts).

3. RESOURCE ALLOCATION:

Direct healthcare resources to the most affected regions: Maharashtra, New York, and Sao Paulo.

4. TREATMENT PROTOCOLS:

Develop age-specific treatment protocols, especially for elderly patients.

5. EARLY WARNING SYSTEMS:

Strengthen surveillance in regions showing rapid case growth for early intervention.

VISUALIZATION OUTPUTS

The following visualizations have been generated as part of this analysis:

Figure 1: demographic analysis

Figure 2: infection patterns

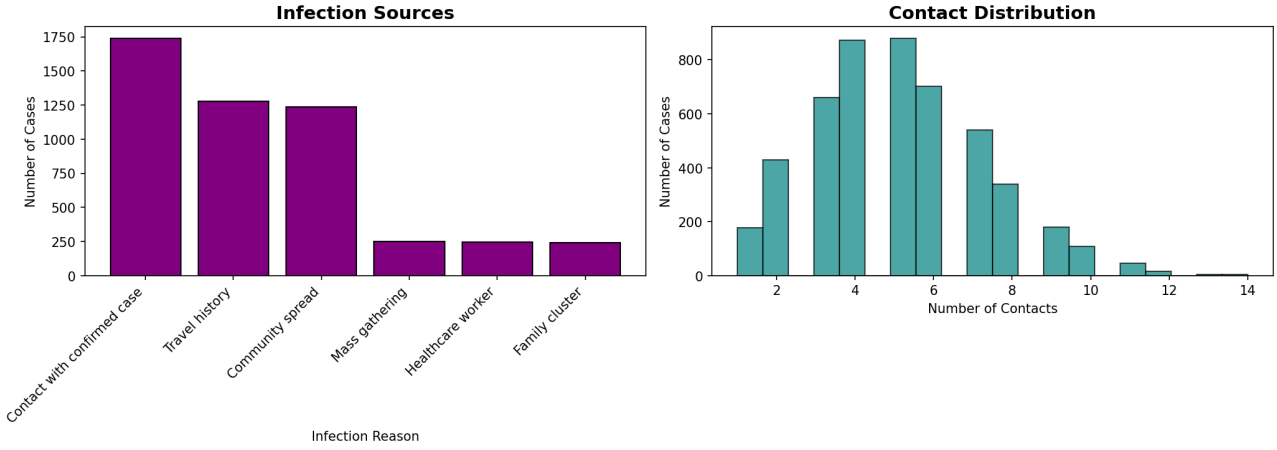


Figure 3: recovery trends

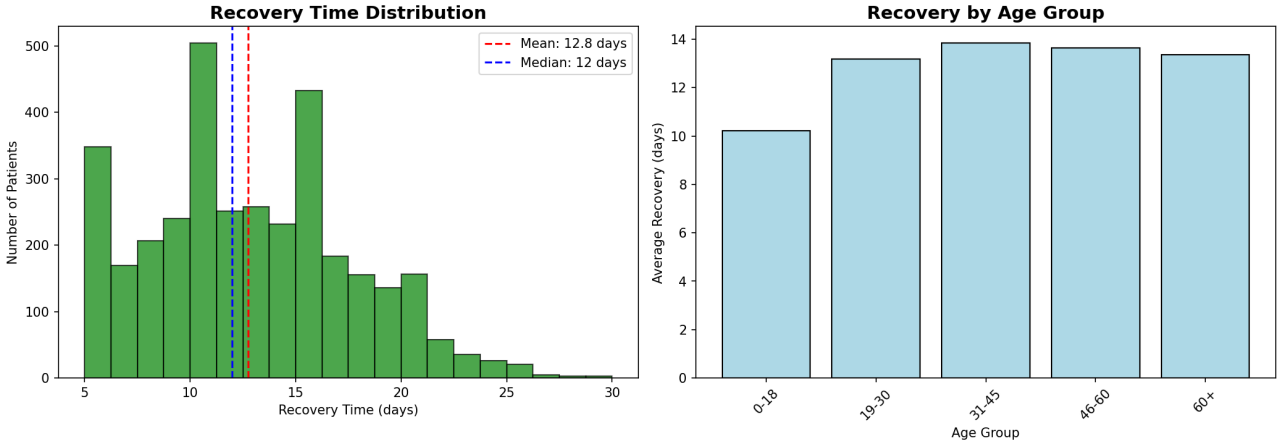


Figure 4: regional impact

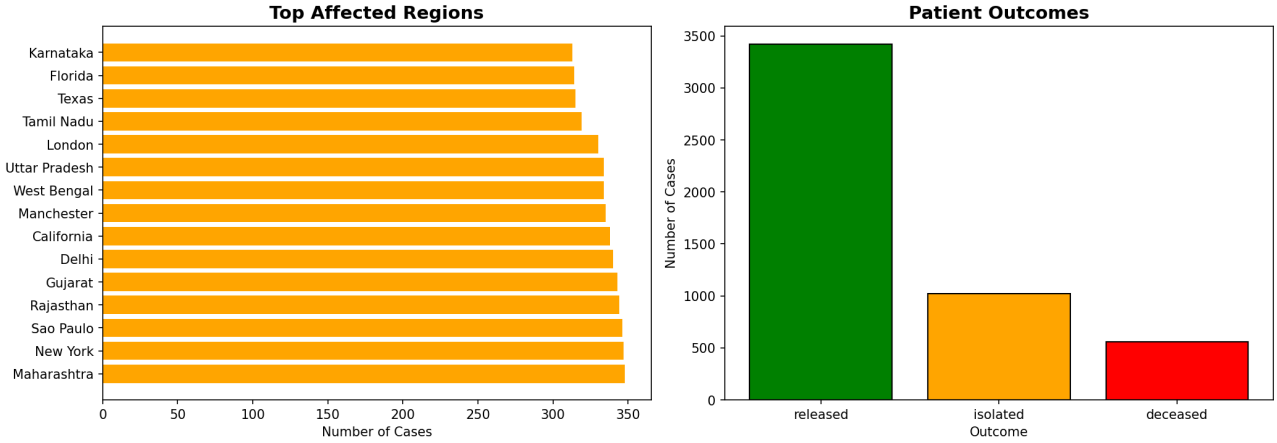


Figure 5: recovery factors

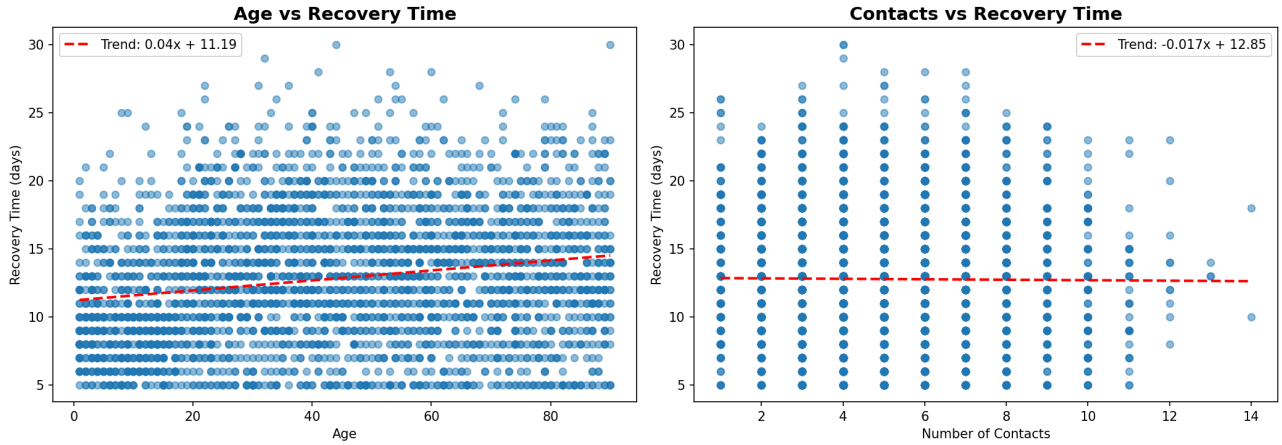
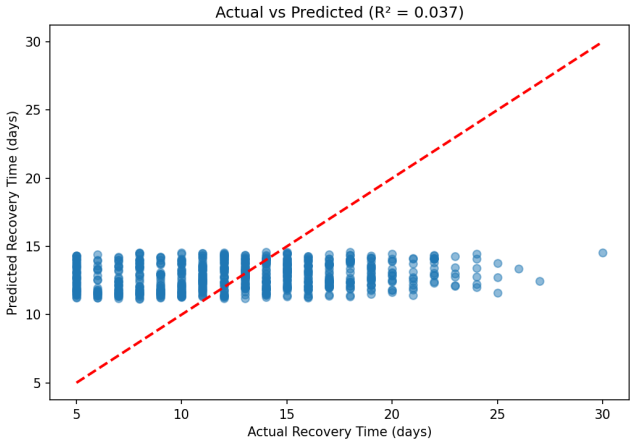
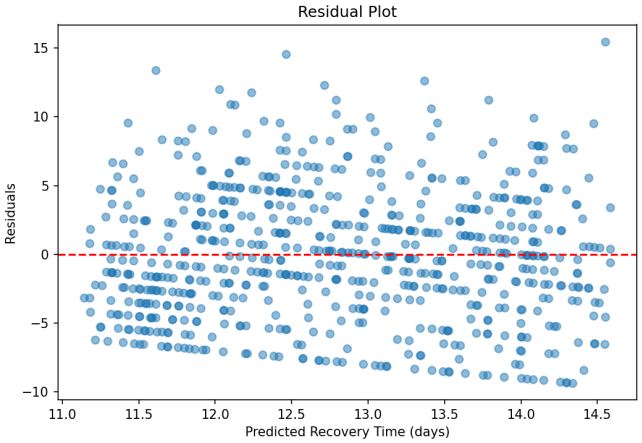


Figure 6: regression model


Actual vs Predicted (R² = 0.037)
Predicted Recovery Time (days)
Actual Recovery Time (days)
30
25
20
15
10
5
5
10
15
20
25
30


Residual Plot
Residuals
Predicted Recovery Time (days)
15
10
5
0
−5
−10
11.0
11.5
12.0
12.5
13.0
13.5
14.0
14.5

TECHNICAL DETAILS

Analysis Tools:
- Programming Language: Python 3.x
- Libraries: Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn
- Statistical Methods: Descriptive statistics, Correlation analysis, Linear Regression

Dataset Information:
- Total Records: 5,000 patient cases
- Time Period: January 2020 - March 2020
- Key Attributes: Demographics, infection details, timelines, outcomes

Model Performance:
- Linear Regression Model trained on age and contact number features
- Model evaluates factors influencing recovery time

CONCLUSION

This analysis successfully provides actionable insights for public health decision-making during the early phase of an infectious disease outbreak. The identified patterns in demographics, transmission, and recovery can help authorities implement targeted interventions and optimize resource allocation.

The complete analysis code, dataset, and visualizations are available for review and further analysis.

For any questions or additional analysis requirements, please contact the HealthGuard Analytics team.